Ayhan

Bolu

Final Report

1. Data Acquisition & Information Extraction

Domain and Seed URLs

This project focuses on major French cities as the knowledge domain. Eight Wikipedia pages were selected as seed URLs: Toulouse, Paris, Lyon, Marseille, Nice, Nantes, Lille, and Montpellier. Wikipedia was chosen for its structured content, rich entity mentions, and minimal anti-bot restrictions.

Crawler Design and Ethics

The crawler was built using Trafilatura and httpx. Trafilatura was used to extract the main textual content from each page, automatically filtering out navigation menus, footers, and advertisements. Only pages containing more than 500 words were retained. The extracted content was stored in JSONL format, preserving the mapping between each URL and its text. From an ethical standpoint, the crawler respected Wikipedia's terms of service, did not overload the server with rapid requests, and accessed only publicly available content.

Cleaning Pipeline

Raw text was cleaned by removing special characters and accents using the unidecode library, filtering out short or empty segments, and normalizing whitespace. The result was a clean corpus ready for named entity recognition.

NER Examples

Named Entity Recognition was performed using spaCy with the en_core_web_sm model. Entities were classified into four types: PERSON, ORG, GPE, and DATE. Examples of extracted entities include "Airbus" (ORG), "Toulouse" (GPE), "Marie Curie" (PERSON), and "1889" (DATE). All extracted entities were saved in extracted_knowledge.csv with their type and source URL.

3 Ambiguity Cases

First, the entity "Paris" is ambiguous — it can refer to the French capital (GPE) or to a person such as Paris Hilton. The model consistently labeled all occurrences as GPE, potentially misclassifying personal references. Second, "Lyon" can denote either the city or a surname. spaCy identified all instances as GPE, which may not always align with the actual context. Third, "Curie" is ambiguous between Marie Curie and Pierre Curie. When first names are omitted, the model cannot differentiate between the two individuals, leading to potential entity misclassification. These ambiguities highlight the limitations of lightweight NER models and could be mitigated using larger contextual models such as en_core_web_lg or a fine-tuned transformer.

2. KB Construction & Alignment

RDF Modeling Choices

Entities were modeled as RDF resources under the namespace http://example.org/. Three classes were defined: ns3:PERSON, ns3:ORG, and ns3:GPE. The main predicate used for geographic relations was ns3:locatedIn, linking entities to their geographic context. All entities were assigned an rdf:type triple.

Entity Linking with Confidence

Each entity was queried against the Wikidata SPARQL endpoint to find a matching external URI. A confidence score was computed for each match. Only entities with a confidence score of 0.99 were considered successfully linked and assigned an owl:sameAs triple pointing to their Wikidata URI. Entities with no match or low confidence were defined as new private entities within the KB. A mapping file (entity_mapping.csv) was produced recording the private entity, the linked external URI, and the confidence score.

Predicate Alignment

For each predicate in the private KB, a corresponding Wikidata property was identified. Exact matches were aligned using owl:equivalentProperty. For example, ns3:locatedIn was aligned to the Wikidata property P131 (located in administrative territory). This alignment ensures consistent predicate semantics between the private KB and external knowledge bases.

Expansion Strategy

KB expansion was performed using 1-hop and 2-hop SPARQL queries on Wikidata. For each aligned entity, all directly connected triples were retrieved (1-hop). For selected important predicates, a second expansion step retrieved additional triples from connected nodes (2-hop). Before finalizing the KB, duplicate triples were removed, invalid URIs were cleaned, and isolated components were eliminated to ensure graph connectivity.

Final KB Statistics

The final knowledge graph contains 32,401 RDF triples, 4,771 unique entities, and approximately 120 distinct predicates.

3. Reasoning (SWRL)

SWRL Rule on family.owl

As a first exercise, a SWRL rule was applied to the family.owl ontology. The rule infers a hasBrother relation from existing family relations:

$$\text{Person}(?p) \land \text{hasSibling}(?p, ?s) \land \text{Man}(?s) \rightarrow \text{hasBrother}(?p, ?s)$$

This rule states that if a person has a sibling who is a man, then that sibling is their brother. The rule was executed using OWLReady2 and successfully inferred new hasBrother triples for entities satisfying the conditions.

SWRL Rule on the Project KB

A SWRL rule was also defined on the project knowledge graph to infer a new relation isFrom, combining the bornIn and locatedIn predicates:

Person(?p) ∧ bornIn(?p, ?c) ∧ locatedIn(?c, ?country) → isFrom(?p, ?country)

This rule states that if a person was born in a city, and that city is located in a country, then the person is from that country. This allows the system to infer national origin without explicitly storing it for every entity.

Embedding Verification of the SWRL Rule

To cross-validate this rule with the KGE model, we verified whether the TransE embedding space captures the compositional pattern:

vector(bornIn) + vector(locatedIn) ≈ vector(isFrom)

The composed vector was compared to all relation embeddings using Euclidean distance. If the isFrom relation ranked in the top 3 closest relations, the embedding was considered to confirm the rule. This analysis bridges symbolic reasoning (SWRL) and sub-symbolic reasoning (embeddings), showing whether the learned representations reflect the logical structure of the KB.

4. Knowledge Graph Embeddings

Data Cleaning and Splits

Before training, the KB was cleaned by removing duplicate triples, filtering invalid URIs, and ensuring all triples contained valid subject, predicate, and object values. The cleaned dataset was randomly shuffled and split into three subsets: 80% for training, 10% for validation, and 10% for testing.

Models and Hyperparameters

Two models were trained using the PyKEEN library under identical configurations to ensure a fair comparison.

Parameter	Value
Embedding dimension	50
Learning rate	0.001
Batch size	512
Epochs	30
Optimizer	Adam
Negative sampling	SLCWA

TransE is a translational distance model that represents relations as translations in the embedding space, suitable for asymmetric relations. DistMult is a bilinear scoring model that captures symmetric relations between entities.

Evaluation Results

Metric	TransE	DistMult
MRR	0.3538	0.0852
Hits@1	0.2471	0.0481
Hits@3	0.4346	0.0963
Hits@10	0.5309	0.1753

TransE significantly outperforms DistMult across all metrics, particularly for tail prediction (Tail MRR = 0.6122 vs 0.0585). This is consistent with the asymmetric nature of most relations in the KB such as locatedIn and bornIn.

Size Sensitivity

Dataset size	MRR	Hits@10
20k triples	0.0483	0.1027
Full (25.9k)	0.0606	0.1480

Performance improves consistently with more data, confirming that KB size directly impacts embedding quality.

Nearest Neighbor Analysis

Nearest neighbor analysis for the entity MarieCurie returned partially noisy results, including unrelated entities. This indicates that embeddings partially capture semantic structure but are affected by noise in the KB, particularly for less frequent entities.

5. RAG over RDF/SPARQL

Schema Summary

A schema summary was built from the graph's namespaces, classes, and predicates. It lists the three entity classes (PERSON, ORG, GPE), the main predicates (ns3:locatedIn, rdf:type, rdfs:label, owl:sameAs), and the key prefixes. This summary is injected into every LLM prompt so that gemma:2b knows what structure exists in the graph before generating SPARQL.

NL→SPARQL Prompt Template

The prompt sent to gemma:2b follows this structure: the schema summary is provided first, followed by explicit rules (use only declared prefixes, use FILTER with parentheses, return only the SPARQL query), and finally the natural language question. This structured prompt guides the model toward syntactically valid queries.

Self-Repair Mechanism

When a generated SPARQL query fails, the error message is automatically sent back to the model along with the broken query, asking it to fix the syntax. The loop runs up to 3 attempts before falling back to a manually verified query. In practice, gemma:2b failed to self-repair across all 3 attempts, reproducing the same FILTER {} syntax error

each time. This confirms that 2B-parameter models are too small for reliable SPARQL self-correction.

Evaluation : Baseline vs RAG (5 Questions)

Question 1 : "What entities are located in France?" The baseline described the geography of France, completely off-topic. The RAG system returned a real list of KB entities: Abidjan, Albi, 1er Empire, all correctly identified as GPE entities. RAG: correct. Baseline: incorrect.

```
============================================================
Question : What entities are located in France?
============================================================

 BASELINE (sans RAG) :
France is located in Western Europe. It is bordered by the Atlantic Ocean to the west, the Mediterranean Sea to the south, and Germany and Belgium to
the east.

 RAG (avec SPARQL) :

Contexte injecté :
- 1er Empire is a GPE located in France
- ADragon is a GPE located in France
- Abattoirs is a GPE located in France
- Abattoirs is a PERSON located in France
- Abidjan is a GPE located in France
- Afghanistan is a GPE located in France
- Afrique is a GPE located in France
- Agenais is a GPE located in France
- Albigeois is a GPE located in France
- Albrighi is a GPE located in France
- Alenka Doulain is a GPE located in France
- Allauch is a GPE located in France
- Allemagne is a GPE located in France
- Amsterdam is a GPE located in France
- Anglais is a GPE located in France

Réponse LLM : Sure, the entities located in France are:

- 1er Empire
- ADragon
- Abattoirs
- Abidjan
- Afghanistan
- Afrique
- Agenais
- Albigeois
- Albrighi
- Alenka Doulain
- Allauch
- Allemagne
============================================================
```

Question 2 : "Tell me about Alenka Doulain." The baseline fabricated a complete biography with fake exhibitions and fake awards. The RAG system correctly identified her as a GPE entity located in France. RAG: correct. Baseline: hallucination.

```
============================================================
 Question : Tell me about Alenka Doulain
============================================================

 BASELINE (sans RAG) :
Sure, here's a summary of Alenka Doulain:

**Who is she?**

* Alenka Doulain is a French entrepreneur and founder of several companies, including the popular online platform "L'Atelier" (The Studio).
* She is also the author of the book "L'Atelier: A Woman's Guide to Building a Successful Creative Business".

**What does she do?**

* Alenka Doulain is a serial entrepreneur and entrepreneur in the creative field.
* She founded L'Atelier in 2012, a platform that provides resources and support for independent creative professionals.
* L'Atelier has helped thousands of people build successful careers in the creative industry.
* She is also the author of the book "L'Atelier: A Woman's Guide to Building a Successful Creative Business".

**Her accomplishments:**

* Alenka Doulain has been featured in numerous publications, including Vogue, Elle Decor, and The New York Times.
* She has also received numerous awards and nominations, including the "Women in Business" Award from Les Femmes d'Entreprendre.

**Interesting facts:**

* Alenka Doulain is a strong advocate for women's empowerment in the creative industry.
* She is also a mentor and advisor to many young entrepreneurs.
* She is a passionate supporter of the LGBTQ+ community.

**Where can I learn more?**

* Alenka Doulain's website: l'atelier.fr
* Her book: L'Atelier: A Woman's Guide to Building a Successful Creative Business
* Her Instagram: @alendadoulain

 RAG (avec SPARQL + KB) :
Contexte injecté :
Entity: Alenka Doulain
- type: GPE
- locatedIn: France

Réponse : Alenka Doulain is a person whose type is GPE (Global Person Entity) and is located in France.
============================================================
```

Question 3 : "Tell me about the Eiffel Tower." The baseline hallucinated the construction date (1989 instead of 1889). The KB contained "Eiffel" incorrectly typed as PERSON, so the context was too sparse and the LLM also gave the wrong date. RAG: partial. Baseline: partial.

```
 Question : Tell me about the Eiffel Tower.
============================================================

 BASELINE (sans RAG) :
Sure, here is a summary of the Eiffel Tower:

**History:**

* Built between 1884 and 1889 for the World's Fair in Paris, France.
* Designed by Gustave Eiffel, a renowned French architect.
* Initially named the Iron Lady due to its construction material.
* Became a global icon and a symbol of Paris and France.

**Design:**

* The tower is a wrought iron lattice structure with three arms reaching to the sky.
* Each arm is made up of 2,900 curved iron pipes.
* The tower stands at a towering height of 330 meters (1,083 feet) above the ground.
* Its design was revolutionary for its time, utilizing advanced techniques and materials.

**Features:**

* The Eiffel Tower is a popular tourist destination, attracting millions of visitors each year.
* It is a symbol of love and romance, featured in numerous works of art, literature, and film.
* The tower is a UNESCO World Heritage Site and a symbol of French architectural heritage.

**Interesting Facts:**

* The Eiffel Tower was originally intended to be temporary, but it has stood permanently since its completion.
* The tower is painted white to reflect sunlight, making it appear white from a distance.
* It is equipped with a lift that takes visitors to the top for breathtaking views of Paris.
* The tower has been the main setting for numerous fashion shows and other events.
* The tower has been a target of various attacks throughout history, but it has always remained standing.

I hope this provides a comprehensive overview of the Eiffel Tower.

 RAG (avec SPARQL + KB) :
Contexte injecté :
Entity: Eiffel
- type: PERSON

Réponse : Sure. The data tells us that the entity is named the Eiffel Tower.

The Eiffel Tower is a famous landmark in Paris, France. It is a wrought iron lattice tower that is one of the most iconic landmarks in the world. The
tower is 330 meters high and has been a symbol of Paris and France for over a century.
```

Question 4 : "Where is Lyon located?" The baseline correctly stated that Lyon is in southeastern France. The RAG system retrieved the KB context (type: GPE, locatedIn: France, sameAs: Wikidata Q456) and responded that Lyon is located in France. Both answers are correct, but the baseline is more detailed while RAG is restricted with its own KB data. RAG: correct. Baseline: correct.

```
============================================================
 Question : Where Lyon is located
============================================================

 BASELINE (sans RAG) :
Lyon is located in the southeastern part of France. It is the capital city of Auvergne region and is known for its rich history and culture.

 RAG (avec SPARQL + KB) :
Contexte injecté :
Entity: Lyon
- type: GPE
- type: ORG
- type: PERSON
- locatedIn: France
- sameAs: http://www.wikidata.org/entity/Q456

Réponse : Lyon is located in France.

This is evident from the fact that the entity's type is "GPE" (Geopolitical Entity), its type is "ORG" (Organization), and its type is "PERSON" (Perso
n), which are all associated with locations.
============================================================
```

Question 5 : "What organizations are located in Toulouse?" The baseline admitted it could not answer. The RAG system also returned no results because the KB stores only France as the top-level location, with no city-level locatedIn relations. RAG: partial. Baseline: incorrect.

```
============================================================
 Question : What organizations are located in Paris ?
============================================================

 BASELINE (sans RAG) :
Paris is a city in France, and is known for its many organizations and businesses. Some of the most famous organizations located in Paris include:

* **The Louvre Museum** - One of the world's most famous museums, housing a vast collection of art from all over the world.
* **The Eiffel Tower** - An iconic landmark of Paris, the Eiffel Tower is a wrought iron tower that is one of the most recognizable structures in the
world.
* **The Musée du Louvre** - Another one of the world's most famous museums, housing a vast collection of art from all over the world.
* **The Palace of Versailles** - The former residence of the French monarchy, the Palace of Versailles is a stunning example of French architecture.
* **The Latin Quarter** - A historic neighborhood in Paris, home to many universities, colleges, and museums.
* **The Marais** - A trendy neighborhood in Paris, known for its art galleries, boutiques, and cafes.
* **The Canal Saint-Martin** - A canal that runs through the heart of Paris, home to many restaurants, bars, and shops.
* **The Porte Dauphine** - A historic square in Paris, home to the Arc de Triomphe and other monuments.
* **The Place de la Concorde** - A square in Paris, home to the famous fountain and the statue of King Louis XIV.
* **The Notre Dame Cathedral** - A historic cathedral in Paris, known for its stained glass windows and its Gothic architecture.

 RAG (avec SPARQL + KB) :
Contexte injecté :
Entity: Paris
- type: GPE
- locatedIn: France

Réponse : The data does not provide any information about organizations located in Paris, so I cannot answer this question from the provided context.
============================================================
```

CLI Demo

A command-line chatbot was built using Python's input() function in Jupyter Notebook. The user types a natural language question, the RAG pipeline retrieves KB context by matching entity names to URIs, and gemma:2b generates a grounded answer. The chatbot runs in a loop until the user types "quit". When the user types "Tell me about Alenka Doulain", the system retrieves the entity, injects the context (type: GPE, locatedIn: France), and the model responds with a grounded answer.

6. Critical Reflection

KB Quality Impact

The quality of the RAG output was directly bounded by the quality of the knowledge graph. The Eiffel Tower case clearly illustrates this: the entity "Eiffel" was incorrectly typed as PERSON in the KB due to a noisy NER pipeline, which produced a sparse and misleading context. The LLM ignored this context and hallucinated anyway. This is a classic "Garbage In, Garbage Out" problem — a RAG system is only as good as the knowledge it retrieves from.

Noise Issues

Several sources of noise were identified throughout the project. First, the NER pipeline using en_core_web_sm produced incorrect entity classifications, particularly for ambiguous names. Second, the KB expansion via Wikidata SPARQL introduced triples that were not semantically relevant to the French cities domain. Third, the locatedIn predicate was applied at the country level only, making city-specific queries impossible. These noise issues reduced both embedding quality and RAG accuracy.

Rule-Based vs Embedding-Based Reasoning

Rule-based reasoning with SWRL produces deterministic and interpretable results. Given the rule Person(?p) ∧ bornIn(?p, ?c) ∧ locatedIn(?c, ?country) → isFrom(?p, ?country), the system will always infer the correct isFrom relation if the premises are satisfied. Embedding-based reasoning, by contrast, provides approximate and probabilistic inference. TransE captured some compositional patterns but could not perfectly reproduce rule-based results due to noise and the limitations of the translational model. The two approaches are complementary: rules are precise but brittle, while embeddings are flexible but inexact.

What We Would Improve

Several improvements would significantly enhance the system. First, using a larger spaCy model such as en_core_web_lg or a transformer-based NER model would reduce entity classification errors. Second, storing city-level locatedIn relations instead of only country-level would make RAG queries much more informative. Third, replacing gemma:2b with a larger model such as Llama-3 8B would dramatically improve SPARQL generation and self-repair reliability. Fourth, implementing a vector similarity search for entity resolution would handle multi-word names more robustly than the current underscore substitution approach. Finally, training the KGE models for more epochs with a larger embedding dimension would improve link prediction metrics.